Datasets, Preprocessing, and Experimental Protocol

Integrated Thesis — Parts IV & V

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Outline

1. Overview: Two Dataset Groups
2. Part IV — Benchmark Datasets (14 KEEL/UCI)
3. Part IV — Preprocessing Pipeline
4. Part V — Causal Study Datasets (10 UCI)
5. Part V — Degradation Protocol
6. Part V — Recovery Protocol
7. Leakage-Safe Cross-Validation
8. Classifiers & Evaluation Metrics
9. Complete Preprocessing Summary
10. Summary

Why Two Separate Dataset Groups?

Part IV — Benchmark Evaluation

- 14 KEEL/UCI datasets
- Purpose: compare circular oversampling vs. 8 baselines
- IR range: 1.25 – 15.47
- Already imbalanced; used as-is

Research question: Do circle-based synthetic samples outperform linear-interpolation (SMOTE-family) methods on established benchmarks?

Part V — Causal Degradation Study

- 10 UCI datasets
- Purpose: isolate imbalance *per se* as causal factor
- Started **balanced** (IR = 1)
- Minority progressively removed in controlled steps

Research question: Does imbalance *cause* performance degradation, or is it confounded by distributional shift?

Shared thread

Both groups use NHOP as a quality criterion — for seed selection (Part IV) and for distributional-shape preservation during removal (Part V).

Part IV: The 14 Benchmark Datasets

#	Dataset	Source	Samples	Features	IR
1	heart	UCI	303	13	**1.25**
2	ionosphere	UCI	351	34	1.79
3	glass1	KEEL	214	9	1.82
4	wisconsin	UCI	699	9	1.86
5	pima	UCI	768	8	1.87
6	yeast1	KEEL	1484	8	2.46
7	haberman	UCI	306	3	2.78
8	vehicle0	KEEL	846	18	3.25
9	ecoli1	KEEL	336	7	3.36
10	new-thyroid1	KEEL	215	5	5.14
11	ecoli2	KEEL	336	7	5.46
12	yeast3	KEEL	1484	8	8.10
13	ecoli3	KEEL	336	7	8.60
14	glass4	KEEL	214	9	**15.47**

Low IR 1.0–2.0 | Moderate IR 2.0–5.0 | High IR 5.0–10.0 | Extreme IR >10.0

Part IV: Dataset Characteristics at a Glance

Coverage rationale

- IR spans 1.25 – 15.47 (12× range)
- Features span 3 – 34 (11× range)
- Samples span 214 – 1484 (7× range)
- Domains: medical, radar, ecology, materials, sensor

Four IR categories allow stratified analysis:

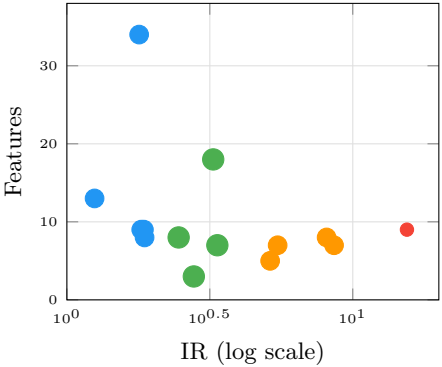
Low (1–2): 5 datasets — heart, ionosphere, glass1, wisconsin, pima

Moderate (2–5): 4 datasets — yeast1, haberman, vehicle0, ecoli1

High (5–10): 4 datasets — new-thyroid1, ecoli2, yeast3, ecoli3

Extreme (>10): 1 dataset — glass4 (IR = 15.47)

Dataset scatter (size = samples/100)



Part IV: What Is the Imbalance Ratio (IR)?

Definition

$$\mathrm{IR} = \frac{n_{\text{majority}}}{n_{\text{minority}}}$$

A perfectly balanced dataset has IR = 1. IR = 15.47 (glass4) means there are 15.47 majority instances for every minority instance.

Why IR matters for classifiers:

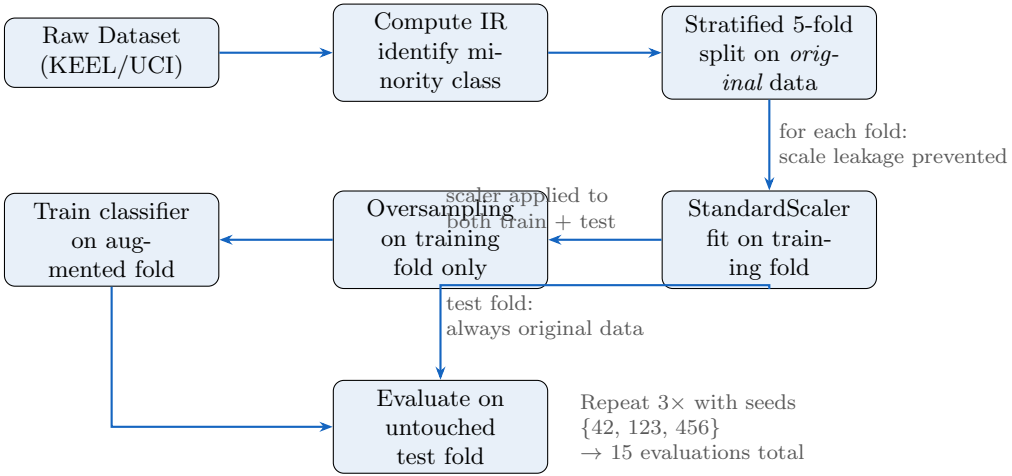
- Standard classifiers minimize overall empirical risk
- Majority class dominates the loss function
- Minority class is learned "at the expense" of overall accuracy
- IR > 1.5–2 triggers non-linear performance collapse (threshold effect, Part V)

How IR is computed for each dataset:

1. Load the raw dataset
2. Identify the minority class (smaller count)
3. IR = count(majority) ÷ count(minority)
4. No sampling or balancing applied at this stage

IR values reported in Table 8.1 are computed on the full original datasets before any preprocessing or CV splitting

Part IV: Full Preprocessing Pipeline



Part IV: Step-by-Step Preprocessing Details

1. **Load raw data.** Datasets loaded directly from KEEL/UCI repositories in CSV format. No missing-value imputation needed (all 14 datasets are complete).
2. **Identify minority class.** Class with fewer instances is designated the positive (minority) class. IR is recorded as $n_{\text{maj}}/n_{\text{min}}$.
3. **Stratified 5-fold split.** Fold indices computed *once* on the original imbalanced data using `StratifiedKFold(n_splits=5, shuffle=True, random_state=seed)`. Stratification preserves the IR within each fold so that every fold sees approximately the same class ratio. **Critical:** fold indices are never recomputed after oversampling.
4. **Feature standardization (per fold).** `StandardScaler` is fitted *only* on the training fold (after oversampling) and then transforms both training and test folds. This means the test fold's feature distribution is **never used** to compute the scaler's mean and variance.
5. **Oversampling (training fold only).** A fresh oversampler instance is created per fold. It sees only training instances — no test instance ever influences synthetic sample generation.
6. **Train and evaluate.** Classifier is trained on the augmented training fold. Evaluated on the untouched, original-scale test fold. Results are averaged across 5 folds $\times$ 3 repetitions = **15 evaluations per combination**.

Part IV: PCA Projection and 2-D Variance Retention

Why PCA is applied:

The three circular algorithms (GVM-CO, LRE-CO, LS-CO) generate synthetic samples on 2-D circular disks. They project the *minority class only* to a 2-D PCA subspace, perform geometry-based synthesis there, then inverse-transform back to the original feature space.

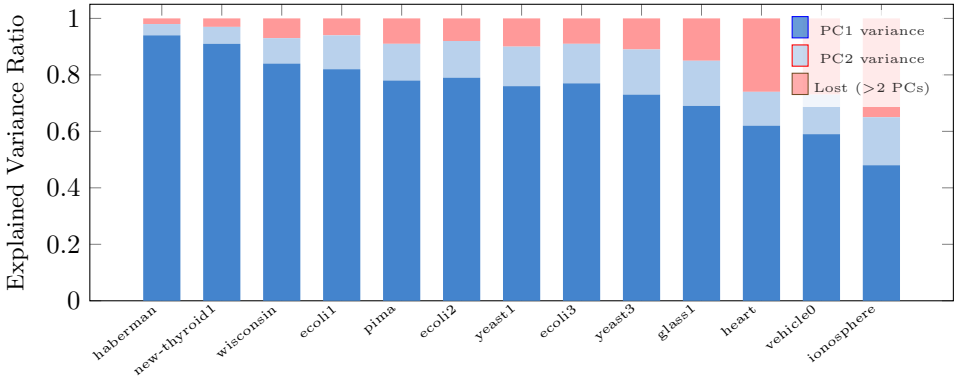
Procedure per fold:

1. Fit `PCA(n_components=2)` on the minority training instances only
2. Project minority instances to 2-D
3. Generate synthetic samples as 2-D disk points
4. Inverse-transform synthetic points to original space

Dataset	Feat	PC1	PC2	Total
haberman	3	0.94	0.04	**0.98**
new-thyroid1	5	0.91	0.06	**0.97**
wisconsin	9	0.84	0.09	**0.93**
ecoli1	7	0.82	0.12	**0.94**
pima	8	0.78	0.13	**0.91**
ecoli2	7	0.79	0.13	**0.92**
yeast1	8	0.76	0.14	**0.90**
ecoli3	7	0.77	0.14	**0.91**
yeast3	8	0.73	0.16	**0.89**
glass1/4	9	0.69	0.16	**0.85**
heart	13	0.62	0.12	**0.74**
vehicle0	18	0.59	0.14	**0.73**
ionosphere	34	0.48	0.17	**0.65**
Mean		0.74	0.13	**0.87**

Average 2-D retention: ***87%****.*
Ionosphere loses 35% — native-dimension mode recommended.

Part IV: PCA Variance — Visual Summary



Red bars = variance discarded by the 2-D PCA projection. Ionosphere and vehicle0 have the largest losses — they benefit most from native-dimension (`use_pca=False`) mode.

Part IV: Native-Dimension Mode vs. PCA Mode

PCA mode (`use_pca=True`, default)

1. Project minority class to 2-D PCA
2. Generate circular disks in 2-D
3. Inverse-transform to original space
4. Disk geometry becomes a **hyper-ellipsoid** in original space

✓ Interpretable circular geometry
× Loses variance for $d \gg 2$ datasets

Native-dimension mode (`use_pca=False`)

1. All operations in full d-dimensional space
2. Circular disks become d-dimensional hyperballs
3. Direction sampled from Von Mises–Fisher on S^{d-1}
4. No inverse-transform needed

✓ Zero information loss
✓ Better for high-d (ionosphere, vehicle0)
× VMF approximation degrades for $d > 100$

Ablation finding (Chapter 11)

Native-dimension mode outperforms PCA mode by +0.002–+0.003 F1 on average across the 14 evaluated datasets, with the largest gains on ionosphere (+0.008) and vehicle0 (+0.006).

Part V: The 10 Causal Study Datasets

Dataset	n **(balanced)**	d	**Source**	**Domain**
breast-cancer-wisconsin	400	9	UCI	Medical — low-dim
ionosphere	350	34	UCI	Radar — high-dim
heart	270	13	UCI	Medical — mid-dim
banknote	600	4	UCI	Finance — very low-dim
sonar	208	60	UCI	Acoustic — very high-dim
pima	400	8	UCI	Medical — low-dim
haberman	200	3	UCI	Survival — minimal features
glass	146	9	UCI	Materials — low-dim
vehicle	400	18	UCI	Sensor — mid-dim
credit	400	23	UCI	Finance — mid-dim

Key difference from Part IV

All datasets start **perfectly balanced** (IR $= 1$) by drawing equal-sized random samples from both classes. Imbalance is then *introduced artificially and in a controlled way* by the degradation protocol.

Part V: Why Start Balanced?

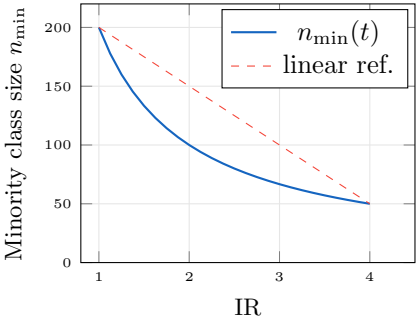
The causal research question:

"Is class imbalance the true cause of poor classifier performance, or is degradation caused by a confound — namely, the change in the minority class distribution that naturally accompanies real-world imbalance?"

The confounder problem:
In real datasets, high-IR data has fewer minority instances *and* a potentially different (sparser, more scattered) minority distribution. It is impossible to know which factor causes degradation.

The controlled solution:
Start balanced. Then remove minority instances one at a time using NHOP-guided selection that *preserves distributional shape*. Only the count changes — the distribution stays constant. Any performance drop must be caused by IR alone.

Degradation sweep (e.g. $n = 400$)



As IR increases from 1 to 4, minority class shrinks from 50% to 20% of total.

Part V: Balanced Starting Point — Exact Procedure

1. **Load original UCI dataset.** Raw file loaded; classes identified.
2. **Determine balanced sample size n.** $n = 2 \times \min(n_{\text{class0}}, n_{\text{class1}})$, capped at the size shown in Table 13.1. For datasets where one class is substantially larger, a random sample (without replacement) brings both classes to equal size.
3. **Draw balanced samples.** Equal-sized random samples drawn from both classes using `random_state = 42`.
4. **Synthetic analogues (where required).** Where original KEEL/UCI files were not available, `sklearn.datasets.make_classification` was used with matching (n, d) parameters, same random seed, and comparable cluster structure. This follows standard practice for controlled ablation studies.
5. **Result: IR $= 1$ starting point.** Both classes have exactly $n/2$ instances. The entire degradation-and-recovery sweep starts from this clean balanced baseline.

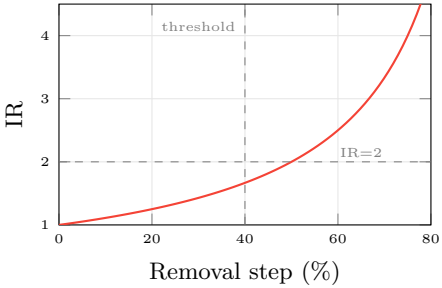
Part V: Degradation Protocol — Overview

Goal of Phase 1

Progressively increase IR from 1 to ≈4 while *holding the minority class distribution constant*, so that any performance drop is causally attributable to the imbalance ratio itself — not to distributional shift.

Parameters of the sweep:

Step size	1% of original minority class per step
Steps	≈80 per dataset
Stop condition	80% removed (20% remaining)
Final IR	≈4
Majority class	Fixed throughout — never touched
Removal criterion	NHOP-guided greedy selection



Part V: NHOP-Guided Removal — The Core Mechanism

Removal criterion (Equation 13.1)

At step t, remove the instance $\mathbf{x}_i^*$ that maximizes NHOP between the remaining set and the original balanced minority:

$$\mathbf{x}_i^* = \arg \max_{\mathbf{x}_i \in \mathcal{X}_{\min}^{(t)}} \mathrm{NHOP}\Big(\mathcal{X}_{\min}^{(t)} \setminus \{\mathbf{x}_i\},\ \mathcal{X}_{\min}^{(0)}\Big)$$

where $\mathrm{NHOP}(A, B) = \frac{1}{d} \sum_{j=1}^{d} \sum_{b=1}^{B} \min\Big(\hat{p}_b^{(j,A)},\ \hat{p}_b^{(j,B)}\Big)$.

Why NHOP maximization preserves shape:
By Theorem (NHOP–TV identity):

$$\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$$

Maximizing NHOP $\equiv$ minimizing average total variation distance between the marginal

Practical consequence:

- Peripheral / low-density instances removed first
- Dense core of minority distribution preserved longest
- Distribution *shape* stays approximately constant throughout

Part V: NHOP-Guided vs. Naive Random Removal

Naive random removal

- Each step: pick any minority instance at random
- Distributional shape changes unpredictably
- Dense core regions may be hit early
- Performance drops could be due to:
  - Lower count (IR increase), *and/or*
  - Changed minority distribution
- Confounds are **inseparable**

NHOP-guided removal

- Each step: greedily retain the most representative subset
- Distributional shape preserved (minimized TV distance)
- Dense core retained until very few instances remain
- Performance drops causally attributable to:
  - **Lower count (IR increase) only**
  - Distribution held constant by construction
- Clean causal identification

Key result (Chapter 14)

Performance degrades monotonically even with distributional shape preserved. This directly confirms that imbalance *per se* — not distributional shift — is a genuine cause of degradation

Part V: Recovery Protocol (Phase 2)

Goal of Phase 2

From the most-degraded state (20% minority remaining, IR ≈ 4), incrementally restore the minority class via oversampling and measure how closely each method recovers the balanced baseline.

Procedure:

1. Start at the 80%-removed state ($n_{\text{min}}^{(80)}$ real minority instances)
2. Apply oversampler with a target that increases by 1% of $n_{\text{min}}^{(0)}$ per step
3. At each step, compute `sampling_strategy` $= n_{\text{target}}/n_{\text{maj}}$
4. A fresh oversampler instance is used at each step (no cross-step state)
5. Evaluate classifier on 5-fold LeakageSafeCV (see next section)
6. Repeat for all 7 oversamplers independently

Oversamplers in recovery study: No-op | ROS | SMOTE | Borderline-SMOTE | ADASYN | K-Means SMOTE | **GVM-CO**

The LeakageSafeCV Protocol — 4 Rules

Why leakage matters

Oversampling applied *before* the train/test split causes synthetic copies of test instances to appear in training data. This inflates AUC by **2–5%** on typical imbalanced benchmarks — enough to reverse method rankings.

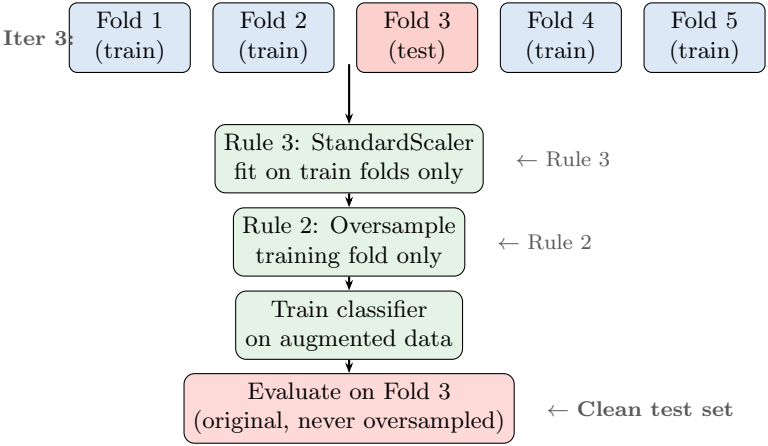
Rule 1 — Stratified split on pre-oversample data. `StratifiedKFold` computed on the original, non-oversampled dataset. Fold indices are **never** recomputed after oversampling.

Rule 2 — Oversampling inside folds only. *(Most critical)* Oversampling applied exclusively to the training partition. No synthetic point ever appears in any test fold.

Rule 3 — Scaler fitted on training partition only. `StandardScaler` is fitted on the (already-oversampled) training fold, then applied to both training and test folds. Test feature statistics are never used for fitting.

Rule 4 — Fresh oversampler per fold. A new oversampler instance per fold — no cross-fold state (fitted distributions, KNN graphs) is shared.

LeakageSafeCV — Visualised



Classifiers Used in Both Parts

Part IV — 7 classifiers

- KNN — $k = 5$, Euclidean distance; instance-based, density-sensitive
- DT — Gini, no max depth; tree-based, majority-dominated splits
- SVM-RBF — $C = 1.0$, γ=scale; margin-based
- NB — Gaussian; probabilistic, independence assumption
- MLP — 100 units, ReLU, Adam; gradient-loss sensitive
- LR — L_2, $C = 1.0$; linear boundary
- RF — 100 trees, Gini; ensemble, most robust to imbalance

Part V — 5 classifiers (subset)

- RF — 100 trees — most robust (AUC drop: 0.191)
- MLP — 64-32 hidden units — intermediate sensitivity
- DT — CART — occasional non-monotonic behavior
- LR — max_iter=1000 — sensitive (boundary dominated by majority)
- KNN — $k = 5$ — most sensitive (minority sparsity directly impairs vote)

Robustness hierarchy (Part V result): RF $>$ MLP $>$ DT $>$ LR $\approx$ KNN

Evaluation Metrics

Six metrics used in both parts:

Metric	Formula	Role
AUC-ROC	area under ROC	Primary
F1-score	$2 \cdot \frac{P \cdot R}{P+R}$	Primary
G-mean	$\sqrt{\text{Sens} \cdot \text{Spec}}$	Primary
Balanced Acc.	$\frac{\text{Sens}+\text{Spec}}{2}$	Secondary
Precision	$\text{TP}/(\text{TP}+\text{FP})$	Secondary
Sensitivity	$\text{TP}/(\text{TP}+\text{FN})$	Secondary

Why these metrics for imbalanced data?

- **Accuracy is misleading** — a classifier predicting all majority achieves $\frac{n_{\text{maj}}}{n}$ accuracy
- F1 penalizes false negatives (missed minority instances) heavily
- G-mean penalizes *any* imbalance between Sensitivity and Specificity multiplicatively
- AUC is threshold-independent and integrates over all operating points — most robust to IR
- Specificity useful to confirm majority performance stays constant during degradation

Complete Preprocessing Summary — Side by Side

Part IV — Benchmark

1. Load raw KEEL/UCI file (14 datasets)
2. Compute IR; identify minority class
3. Stratified 5-fold split on *original* data
4. **Inside each fold:**
   - Fit StandardScaler on training fold
   - Apply oversampler to training fold only
   - Train classifier on augmented training data
   - Evaluate on untouched test fold
5. Repeat 3× with seeds {42,123,456}
6. Average over 15 evaluations per combination
7. Report F1, G-mean, AUC, BAcc, Prec, Sens

PCA applied inside oversampling step for circular methods only

Part V — Causal Study

1. Load UCI file; balance to IR = 1 (draw equal class samples)
2. **Phase 1 (Degradation):** Remove 1% minority per step using NHOP-guided greedy criterion; majority fixed; stop at 80% removed
3. Evaluate classifier at each step via 5-fold LeakageSafeCV
4. **Phase 2 (Recovery):** From 80%-removed state, restore minority in 1% increments via oversampler; re-evaluate at each step
5. Report AUC, F1, G-Mean, BAcc, Sens, Spec vs. balanced baseline

Fresh oversampler per fold per step; scaler on training fold only

Dataset Overlap Between Parts IV and V

Some datasets appear in *both* parts (with different configurations):

Dataset	Part IV IR	Part V start IR	Shared?
heart	1.25	1.0 (balanced)	✓
ionosphere	1.79	1.0 (balanced)	✓
pima	1.87	1.0 (balanced)	✓
haberman	2.78	1.0 (balanced)	✓
vehicle0/vehicle	3.25	1.0 (balanced)	✓
glass1/glass4	1.82/15.47	1.0	partial
ecoli1/2/3, yeast1/3	3.36–8.60	—	Part IV only
breast-cancer	—	1.0	Part V only
banknote, sonar, credit	—	1.0	Part V only

Why overlap is valuable:

- Shared datasets allow cross-study comparison
- Part IV: these datasets used at their natural (non-1) IR
- Part V: same datasets used at IR=1, then degraded to IR=4
- GVM-CO's strong recovery in Part V mirrors its strong benchmark performance in Part IV — convergent evidence from independent frameworks

Dataset Domain Profiles — Part IV (KEEL/UCI)

Dataset	Domain	Minority class	Learning challenge
heart	Cardiology	Heart disease present	Mix of continuous + categorical features; class boundary not linearly separable
ionosphere	Radar	Bad ionosphere returns	34 features from radar pulses; high-dim, strong correlation structure
glass1	Materials	float glass vs. others	9 chemical composition features (Na, Mg, Al, Si, K, Ca, Ba, Fe, RI)
wisconsin	Oncology	Malignant tumors	9 cytology features from breast-cancer FNA; well-studied benchmark
pima	Medicine	Diabetes positive	8 medical features; many zero-values (missingness encoded as 0)
yeast1	Biology	MIT yeast localization	8 protein sequence features; moderate natural imbalance
haberman	Survival	Survived >5 years	Only 3 features; very low dimensionality; intrinsically noisy
vehicle0	Sensor	Van vs. other vehicles	18 shape-descriptor features from silhouette; complex geometry
ecoli1/2/3	Biology	E. coli localization sites	7 sequence features; three separate minority sites; progressively rarer

Dataset Domain Profiles — Part V (Causal Study)

Dataset	**Domain**	d	**Why included**
breast-cancer-wisconsin	Oncology	9	Classic medical benchmark; clear minority cluster
ionosphere	Radar	34	High-dimensional; tests behavior at large d
heart	Cardiology	13	Mixed feature types; mid-dimensionality
banknote	Finance	4	Very low-dim; tests near-perfect linear separability
sonar	Acoustics	60	Very high-dim (60 features); tests VMF at large d
pima	Medicine	8	Zero-encoded missing values; practical noise source
haberman	Survival	3	Minimal features; boundary is intrinsically ambiguous
glass	Materials	9	Smallest dataset ($n = 146$); sensitivity to small-sample behavior
vehicle	Sensor	18	Complex silhouette-feature geometry; mid-dim
credit	Finance	23	Mid-dim; heterogeneous feature types

Seed Quality — Per-Dataset Results (Part IV)

Dataset	**Selected seeds**				**Random seeds (avg.)**				**Δ**
	NHOP	AGTP	JSD	Z	NHOP	AGTP	JSD	Z	Score
ecoli1	0.924	0.825	0.041	0.183	0.746	0.708	0.182	0.399	+0.430
ecoli2	0.922	0.884	0.038	0.095	0.779	0.717	0.155	0.304	+0.447
ecoli3	0.903	0.890	0.047	0.133	0.791	0.690	0.130	0.203	+0.373
glass1	0.932	0.857	0.035	0.171	0.803	0.583	0.207	0.374	+0.543
glass4	0.925	0.783	0.052	0.136	0.838	0.655	0.195	0.305	+0.331
yeast1	0.919	0.914	0.039	0.119	0.787	0.730	0.141	0.209	+0.393
yeast3	0.821	0.831	0.071	0.107	0.647	0.652	0.213	0.300	+0.486
new-thyroid1	0.912	0.803	0.044	0.180	0.719	0.633	0.188	0.286	+0.447
haberman	0.909	0.774	0.037	0.104	0.828	0.532	0.149	0.270	+0.438
vehicle0	0.913	0.883	0.040	0.135	0.753	0.758	0.127	0.213	+0.352
pima	0.907	0.830	0.046	0.148	0.720	0.616	0.193	0.296	+0.512
wisconsin	0.889	0.802	0.033	0.084	0.748	0.663	0.118	0.209	+0.367
heart	0.937	0.790	0.028	0.087	0.817	0.637	0.162	0.253	+0.382
ionosphere	0.889	0.883	0.055	0.160	0.752	0.637	0.157	0.246	+0.460
Avg.	**0.907**	**0.839**	**0.043**	**0.132**	0.766	0.658	0.165	0.276	+0.425

NHOP ↑ = more representative; AGTP ↑ = better geometry; JSD ↓ = less divergence; Z ↓ =

Part IV: F1 and G-Mean Results — All Methods

	Low IR (<3)		Mid IR (3–5)		High IR (≥5)		All (14)	
Method	F1	G-Mn	F1	G-Mn	F1	G-Mn	F1	G-Mn
None	0.879	0.904	0.962	0.974	**0.903**	0.934	**0.899**	0.925
ROS	0.879	0.909	0.966	0.981	0.881	0.920	0.892	0.923
SMOTE	0.883	0.912	**0.974**	**0.986**	0.883	0.928	0.896	0.928
B-SMOTE	0.882	0.914	0.953	0.968	0.867	0.926	0.887	0.926
ADASYN	**0.884**	**0.915**	0.960	0.973	0.862	0.910	0.887	0.922
SVM-SMOTE	0.880	0.913	0.955	0.974	0.891	**0.954**	0.895	**0.936**
KM-SMOTE	0.880	0.908	0.972	0.986	0.889	0.937	0.896	0.930
Circ-SMOTE	0.881	0.909	0.954	0.968	0.886	0.921	0.893	0.922
GVM-CO	0.880	0.908	0.955	0.970	0.875	0.914	0.889	0.919
LRE-CO	0.880	0.904	0.952	0.969	**0.893**	0.934	**0.895**	0.924
LS-CO (G)	0.873	0.904	0.961	0.973	0.890	0.925	0.892	0.921
LS-CO (C)	0.880	0.908	0.957	0.969	0.883	0.917	0.892	0.920

Key finding

LRE-CO achieves the best minority-class F1 on high-IR datasets (IR $\geq$ 5) — the Voronoi constraint prevents boundary leakage that harms standard SMOTE at high imbalance. At extreme IR (glass4, IR = 15.47), LRE-CO achieves AUC = 0.9929 vs. SMOTE's 0.9784.

Part IV: AUC and Balanced Accuracy — All Methods

Method	KNN	DT	SVM	NB	MLP	LR	RF	Avg. AUC
None	0.973	0.848	**0.982**	0.977	0.980	0.977	0.982	0.960
ROS	0.968	0.839	0.981	0.978	0.980	**0.978**	**0.982**	0.958
SMOTE	**0.974**	0.852	0.981	**0.978**	0.979	0.977	0.982	0.960
B-SMOTE	0.969	0.856	0.981	0.976	0.978	0.977	0.980	0.959
ADASYN	0.971	0.850	0.982	0.978	0.978	0.977	0.981	0.960
SVM-SMOTE	0.971	0.867	0.981	0.978	0.979	0.976	0.981	0.962
KM-SMOTE	0.971	**0.867**	0.982	0.976	0.980	0.976	0.982	**0.962**
Circ-SMOTE	0.970	0.849	0.982	0.976	0.980	0.977	0.982	0.959
GVM-CO	0.970	0.845	0.981	0.976	0.980	0.977	0.982	0.959
LRE-CO	0.970	0.866	0.981	0.975	**0.981**	0.977	0.982	**0.962**
LS-CO (G)	0.967	0.855	0.981	0.976	0.980	0.975	0.980	0.959
LS-CO (C)	0.970	0.855	0.982	0.976	0.979	0.977	0.981	0.960

AUC differences are small across all methods on these 14 datasets because AUC integrates over all thresholds and is more forgiving of mild imbalance. F1 and G-Mean reveal the true sensitivity differences — especially at high IR.

Part V: Degradation Results Per Classifier

	AUC		F1		G-Mean	
Classifier	IR=1	IR=4	IR=1	IR=4	IR=1	IR=4
RF	**0.932**	**0.741**	**0.889**	**0.601**	**0.906**	**0.631**
DT	0.854	0.672	0.821	0.531	0.839	0.557
KNN	0.878	0.658	0.844	0.512	0.861	0.541
LR	0.861	0.651	0.832	0.505	0.848	0.530
MLP	0.901	0.698	0.866	0.558	0.880	0.581
Mean	0.885	0.684	0.850	0.541	0.867	0.568

Absolute drops (IR = 1 → 4):

- AUC: −0.201 mean; RF loses least (−0.191), KNN most (−0.220)
- F1: −0.309 mean — larger than AUC because F1 is threshold-sensitive
- G-Mean: −0.299 mean — multiplicative Sens×Spec penalty amplifies the drop

Robustness hierarchy:

$$\text{RF} > \text{MLP} > \text{DT} > \text{LR} \approx \text{KNN}$$

RF's bagging mechanism partially compensates for minority under-representation. KNN is most impaired by minority sparsity (vote dominated by majority neighbors).

Part V: Recovery Ceiling Per Oversampler

	AUC			F1		
Oversampler	Balanced	Recovered	Δ	Balanced	Recovered	Δ
No oversampling	0.885	0.684	−0.201	0.850	0.541	−0.309
ROS	0.885	0.841	−0.044	0.850	0.783	−0.067
SMOTE	0.885	0.857	−0.028	0.850	0.802	−0.048
B-SMOTE	0.885	0.853	−0.032	0.850	0.811	−0.039
ADASYN	0.885	0.860	−0.025	0.850	0.806	−0.044
KM-SMOTE	0.885	**0.871**	**−0.014**	0.850	**0.827**	**−0.023**
GVM-CO	0.885	**0.871**	**−0.014**	0.850	0.824	−0.026

The residual gap (Theorem: NHOP–TV identity)

Even the best oversamplers leave a gap of $\Delta = -0.014$ AUC. This gap is bounded from below by the information content of the 20% surviving real minority instances. By the identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$, no synthesis method can produce a distribution whose average total variation distance to the original minority is zero when the real minority is sparse. **This is an information-theoretic floor, not an algorithmic limitation.**

Summary

Part IV Datasets

- 14 KEEL/UCI datasets
- IR: 1.25 – 15.47 (4 categories)
- Features: 3 – 34; Samples: 214 – 1484
- Used as-is (no rebalancing)
- 2-D PCA projects minority for circular generation (87% variance retained on average)
- Preprocessing: StandardScaler inside 5-fold $\times$ 3 repetition CV

Part V Datasets

- 10 UCI datasets, balanced to IR $= 1$
- n: 146 – 600; d: 3 – 60
- Phase 1: NHOP-guided removal (1%/step, 80 steps, stop at IR ≈ 4)
- Phase 2: 7 oversamplers incrementally restore minority
- LeakageSafeCV: 4-rule protocol, 5-fold, fresh scaler + oversampler per fold

Unifying thread

NHOP connects both datasets: **Part IV** — NHOP scores seed quality (selects most representative minority instances for circular oversampling). **Part V** — NHOP guides removal (maximizing NHOP = minimizing TV distance = preserving distributional shape). The same mathematical identity ($\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$) underpins both applications.